AERIS-10 — MAIN_BOARD — Schematic sheet 1/4

Drawing ID: ELEC-SCH-MB-S1 Revision: A Date: 2026-10-09 Units: mm Scale: 1:1 at the stated page size

Source: 4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.sch (EAGLE 7.4.0 XML; file revision: none recorded in the source) Status: SOURCE-DERIVED

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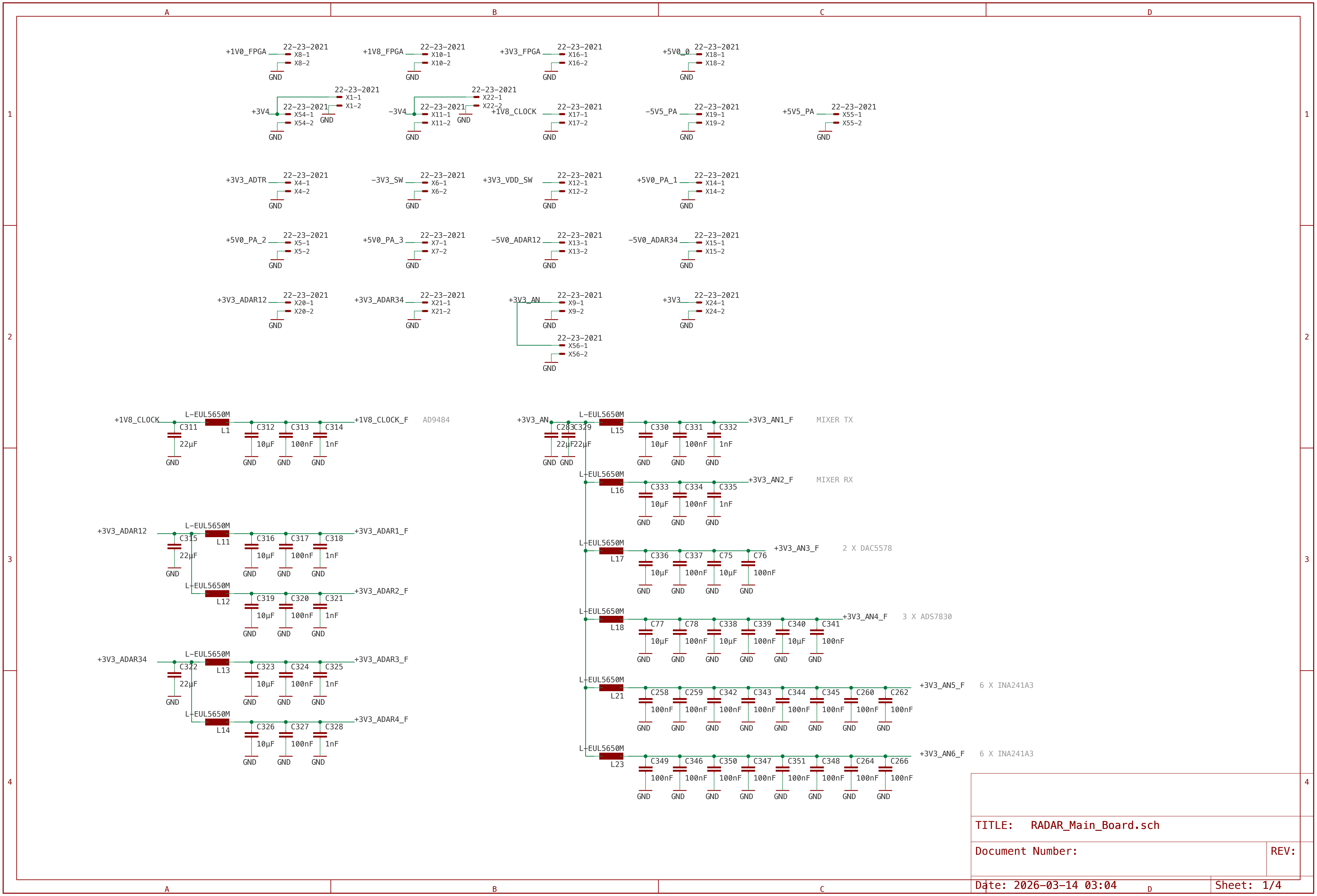
A
B
C
D
1
2
3
4
+1V0_FPGA
22-23-2021
X8-1
X8-2
GND
+1V8_FPGA
22-23-2021
X10-1
X10-2
GND
+3V3_FPGA
22-23-2021
X16-1
X16-2
GND
+5V0_0
22-23-2021
X18-1
X18-2
GND
+3V4
22-23-2021
X1-1
X1-2
GND
22-23-2021
X54-1
X54-2
GND
-3V4
22-23-2021
X22-1
X22-2
GND
22-23-2021
X11-1
X11-2
GND
+1V8_CLOCK
22-23-2021
X17-1
X17-2
GND
-5V5_PA
22-23-2021
X19-1
X19-2
GND
+5V5_PA
22-23-2021
X55-1
X55-2
GND
+3V3_ADTR
22-23-2021
X4-1
X4-2
GND
-3V3_SW
22-23-2021
X6-1
X6-2
GND
+3V3_VDD_SW
22-23-2021
X12-1
X12-2
GND
+5V0_PA_1
22-23-2021
X14-1
X14-2
GND
+5V0_PA_2
22-23-2021
X5-1
X5-2
GND
+5V0_PA_3
22-23-2021
X7-1
X7-2
GND
-5V0_ADAR12
22-23-2021
X13-1
X13-2
GND
-5V0_ADAR34
22-23-2021
X15-1
X15-2
GND
+3V3_ADAR12
22-23-2021
X20-1
X20-2
GND
+3V3_ADAR34
22-23-2021
X21-1
X21-2
GND
+3V3_AN
22-23-2021
X9-1
X9-2
GND
22-23-2021
X56-1
X56-2
GND
+3V3
22-23-2021
X24-1
X24-2
GND
+1V8_CLOCK
L-EUL5650M
L1
C311 22µF
C312 10µF
C313 100nF
C314 1nF
+1V8_CLOCK_F
AD9484
+3V3_ADAR12
L-EUL5650M
L11
C315 22µF
C316 10µF
C317 100nF
C318 1nF
+3V3_ADAR1_F
L-EUL5650M
L12
C319 10µF
C320 100nF
C321 1nF
+3V3_ADAR2_F
+3V3_ADAR34
L-EUL5650M
L13
C322 22µF
C323 10µF
C324 100nF
C325 1nF
+3V3_ADAR3_F
L-EUL5650M
L14
C326 10µF
C327 100nF
C328 1nF
+3V3_ADAR4_F
+3V3_AN
C283 22µF
C329 22µF
L-EUL5650M
L15
C330 10µF
C331 100nF
C332 1nF
+3V3_AN1_F
MIXER TX
L-EUL5650M
L16
C333 10µF
C334 100nF
C335 1nF
+3V3_AN2_F
MIXER RX
L-EUL5650M
L17
C336 10µF
C337 100nF
C75 10µF
C76 100nF
+3V3_AN3_F
2 X DAC5578
L-EUL5650M
L18
C77 10µF
C78 100nF
C338 10µF
C339 100nF
C340 10µF
C341 100nF
+3V3_AN4_F
3 X ADS7830
L-EUL5650M
L21
C258 100nF
C259 100nF
C342 100nF
C343 100nF
C344 100nF
C345 100nF
C260 100nF
C262 100nF
+3V3_AN5_F
6 X INA241A3
L-EUL5650M
L23
C349 100nF
C346 100nF
C350 100nF
C347 100nF
C351 100nF
C348 100nF
C264 100nF
C266 100nF
+3V3_AN6_F
6 X INA241A3
GND

TITLE: RADAR_Main_Board.sch

Document Number:

REV:

Date: 2026-03-14 03:04

Sheet: 1/4

AERIS-10 — MAIN_BOARD — Schematic sheet 2/4

Drawing ID: ELEC-SCH-MB-S2 Revision: A Date: 2026-10-09 Units: mm Scale: 1:1 at the stated page size

Source: 4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.sch (EAGLE 7.4.0 XML; file revision: none recorded in the source) Status: SOURCE-DERIVED

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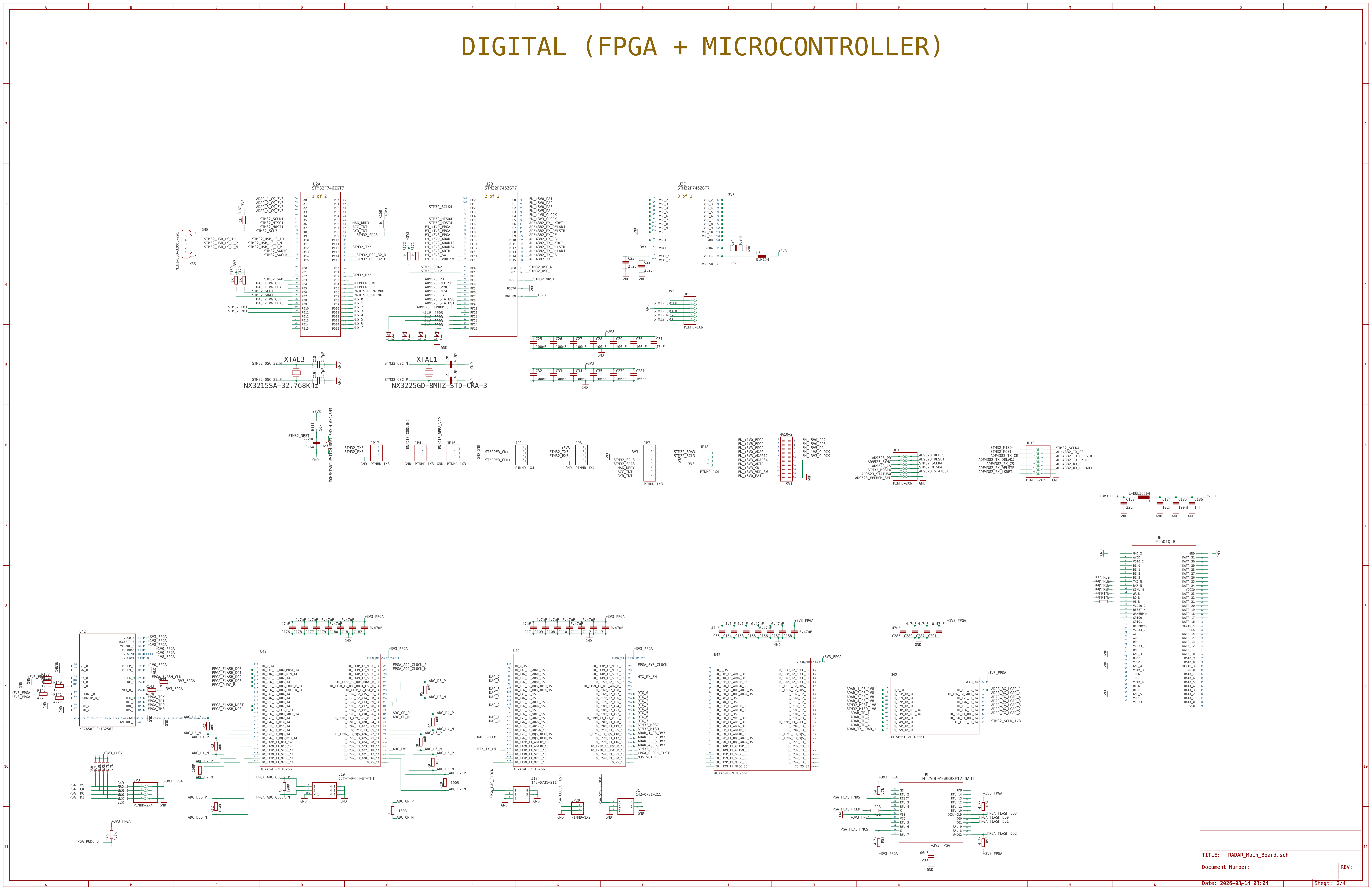
DIGITAL (FPGA + MICROCONTROLLER)


U2A
STM32F746ZGT7
1 of 2
ADAR_1_CS_3V3
ADAR_2_CS_3V3
ADAR_3_CS_3V3
ADAR_4_CS_3V3
STM32_SCLK1
STM32_MISO1
STM32_MOSI1
STM32_SCL3
STM32_USB_FS_ID
STM32_USB_FS_D_N
STM32_USB_FS_D_P
STM32_SWDIO
STM32_SWCLK
STM32_SWO
DAC_1_VG_CLR
DAC_1_VG_LDAC
STM32_SCL1
STM32_SDA1
DAC_2_VG_CLR
DAC_2_VG_LDAC
STM32_TX3
STM32_RX3
MAG_DRDY
ACC_INT
GYR_INT
STM32_SDA3
STM32_TX5
STM32_OSC_32_N
STM32_OSC_32_P
STM32_RX5
STEPPER_CW+
STEPPER_CLK+
EN/DIS_RFPA_VDD
EN/DIS_COOLING
DIG_0
DIG_1
DIG_2
DIG_3
DIG_4
DIG_5
DIG_6
DIG_7
R167 1k +3V3
R168 1k +3V3
R169 1k
R170 1k
MINI-USB-32005-201
X53
GND
U2B
STM32F746ZGT7
2 of 2
STM32_SCLK4
STM32_MISO4
STM32_MOSI4
EN_+1V0_FPGA
EN_+1V8_FPGA
EN_+3V3_FPGA
EN_+5V0_ADAR
EN_+3V3_ADAR12
EN_+3V3_ADAR34
EN_+3V3_ADTR
EN_+3V3_SW
EN_+3V3_VDD_SW
STM32_SDA2
STM32_SCL2
AD9523_PD
AD9523_REF_SEL
AD9523_SYNC
AD9523_RESET
AD9523_CS
AD9523_STATUS0
AD9523_STATUS1
AD9523_EEPROM_SEL
R110 500R
R112 500R
R113 500R
R114 500R
R172 1k
R171 1k
EN_+5V0_PA1
EN_+5V0_PA2
EN_+5V0_PA3
EN_+5V5_PA
EN_+1V8_CLOCK
EN_+3V3_CLOCK
ADF4382_RX_LKDET
ADF4382_RX_DELADJ
ADF4382_RX_DELSTR
ADF4382_RX_CE
ADF4382_RX_CS
ADF4382_TX_LKDET
ADF4382_TX_DELSTR
ADF4382_TX_DELADJ
ADF4382_TX_CS
ADF4382_TX_CE
STM32_OSC_N
STM32_OSC_P
NRST
STM32_NRST
BOOT0
PDR_ON
+3V3
U2C
STM32F746ZGT7
3 of 3
VSS_2 … VSS_9
VSS
VSSA
VBAT
VCAP_1
VCAP_2
VDD_2 … VDD_11
VDD
VDDA
VREF+
VDDUSB
C23 2.2uF
C22 2.2uF
C24 100nF
L5 BLM15H
JP2
STM32_SWCLK
STM32_SWDIO
STM32_NRST
STM32_SWO
PINHD-1X6
C25 100nF
C26 100nF
C27 100nF
C28 100nF
C29 100nF
C30 100nF
C31 47nF
C32 100nF
C33 100nF
C34 100nF
C35 100nF
C279 100nF
C281 100nF
XTAL3
C18 2.7pF
C19 2.7pF
NX3215SA-32.768KHz
XTAL1
C20 4.3pF
C21 4.3pF
NX3225GD-8MHZ-STD-CRA-3
R111 10k
C164 2.2uF
S1
MOMENTARY-SWITCH-SPST-SMD-4.6X2.8MM
JP17 PINHD-1X3
JP4 PINHD-1X3
JP10 PINHD-1X3
JP9 PINHD-1X4
JP8 PINHD-1X4
STM32_TX5
STM32_RX5
JP7 PINHD-1X8
STM32_SCL3
STM32_SDA3
JP18 PINHD-1X4
MA10-2
SV1
JP1 PINHD-2X6
JP13 PINHD-2X7
+3V3_FPGA
L-EUL5650M
L19
C159 22µF
C184 10µF
C185 100nF
C186 1nF
+3V3_FT
U6
FT601Q-B-T
GND_2
AVDD
VD10_2
BE_0
BE_1
BE_2
BE_3
TXE_N
RXF_N
SIWU_N
WR_N
RD_N
OE_N
VCCIO_2
RESET_N
WAKEUP_N
GPIO0
GPIO1
RESERVED
VCC33_2
XI
XO
DP
VCC33_3
DM
GND_3
RREF
VDDA
GND_4
VD10_3
TODN
TODP
VD10_4
RIDN
RIDP
GND_5
VBUS
VCC33
DATA_31 … DATA_0
VCCIO
VCCIO_4
VCCIO_3
CLK
VD10
DV10
33R R60
R61
R83
R84
R145
R146
U42
XC7A50T-2FTG256I
VCCO_0
VCCBATT_0
VCCADC_0
VCCBRAM
VCCINT
VCCAUX
+1V8_FPGA
+1V0_FPGA
VREFP_0
VREFN_0
VP_0
VN_0
M0_0
M1_0
M2_0
CFGBVS_0
PROGRAM_B_0
DXP_0
DXN_0
CCLK_0
DONE_0
INIT_B_0
TCK_0
TDI_0
TDO_0
TMS_0
GNDADC_0
FPGA_FLASH_CLK
FPGA_TCK
FPGA_TDI
FPGA_TDO
FPGA_TMS
R138 1k
R139 1k
R140 1k
R141 4.7k
R142 4.7k
R143 4.7k
R144 1k
R68 4.7k
R67
R66
R65
R49 22R
R62
R63
R64
JP3 PINHD-2X4
R69 4.7k
FPGA_PUDC_B
C176 47uF
C178 4.7uF
C177 4.7uF
C179 0.47uF
C180 0.47uF
C181 0.47uF
C182 0.47uF
FPGA_FLASH_DQ0
FPGA_FLASH_DQ1
FPGA_FLASH_DQ2
FPGA_FLASH_DQ3
FPGA_FLASH_NRST
FPGA_FLASH_NCS
FPGA_ADC_CLOCK_P
FPGA_ADC_CLOCK_N
ADC_D0_P
ADC_D0_N
ADC_D1_P
ADC_D1_N
ADC_D2_P
ADC_D2_N
ADC_D3_P
ADC_D3_N
ADC_D4_P
ADC_D4_N
ADC_D5_P
ADC_D5_N
ADC_D6_P
ADC_D6_N
ADC_D7_P
ADC_D7_N
ADC_OR_P
ADC_OR_N
ADC_PWRD
ADC_DCO_P
ADC_DCO_N
R2 100R
R3 100R
R4 100R
R5 100R
R6 100R
R7 100R
R8 100R
R9 100R
R10 100R
R11 100R
R12 100R
J19
CJT-T-P-HH-ST-TH1
MH1
MH2
MH3
MH4
C17 47uF
C109 4.7uF
C108 4.7uF
C110 0.47uF
C111 0.47uF
C112 0.47uF
C113 0.47uF
DAC_0 … DAC_7
DAC_SLEEP
MIX_TX_EN
FPGA_SYS_CLOCK
MIX_RX_EN
STM32_MOSI1
STM32_MISO1
STM32_SCLK1
FPGA_CLOCK_TEST
M3S_VCTRL
FPGA_DAC_CLOCK
J18
142-0731-211
FPGA_CLOCK_TEST
JP20 PINHD-1X2
J1
142-0731-211
C55 47uF
C154 4.7uF
C153 4.7uF
C155 0.47uF
C156 0.47uF
C157 0.47uF
C158 0.47uF
VCCO_35
C285 47uF
C289 4.7uF
C287 4.7uF
C291 0.47uF
VCCO_34
ADAR_1_CS_1V8
ADAR_2_CS_1V8
ADAR_3_CS_1V8
ADAR_4_CS_1V8
STM32_MOSI_1V8
STM32_MISO_1V8
ADAR_TR_1
ADAR_TR_2
ADAR_TR_3
ADAR_TR_4
ADAR_TX_LOAD_1
ADAR_RX_LOAD_1
ADAR_RX_LOAD_4
ADAR_TX_LOAD_4
ADAR_RX_LOAD_3
ADAR_TX_LOAD_3
ADAR_RX_LOAD_2
ADAR_TX_LOAD_2
STM32_SCLK_1V8
U9
MT25QL01GBBB8E12-0AUT
NC
RFU_2
RESET
RFU_3
RFU_4
C
VSS
VCC
RFU_5
RFU_6
S
RFU_7
RFU
RFU_14
RFU_13
RFU_12
RFU_11
RFU_10
DQ3/HOLD
DQ0
DQ1
RFU_9
RFU_8
W/DQ2
R50 4.7k
R51 22R
R52 4.7k
R53 4.7k
R54 4.7k
C50 100nF
GND

TITLE: RADAR_Main_Board.sch

Document Number:

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Date: 2026-03-14 03:04

Sheet: 2/4

AERIS-10 — MAIN_BOARD — Schematic sheet 3/4

Drawing ID: ELEC-SCH-MB-S3 Revision: A Date: 2026-10-09 Units: mm Scale: 1:1 at the stated page size

Source: 4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.sch (EAGLE 7.4.0 XML; file revision: none recorded in the source) Status: SOURCE-DERIVED

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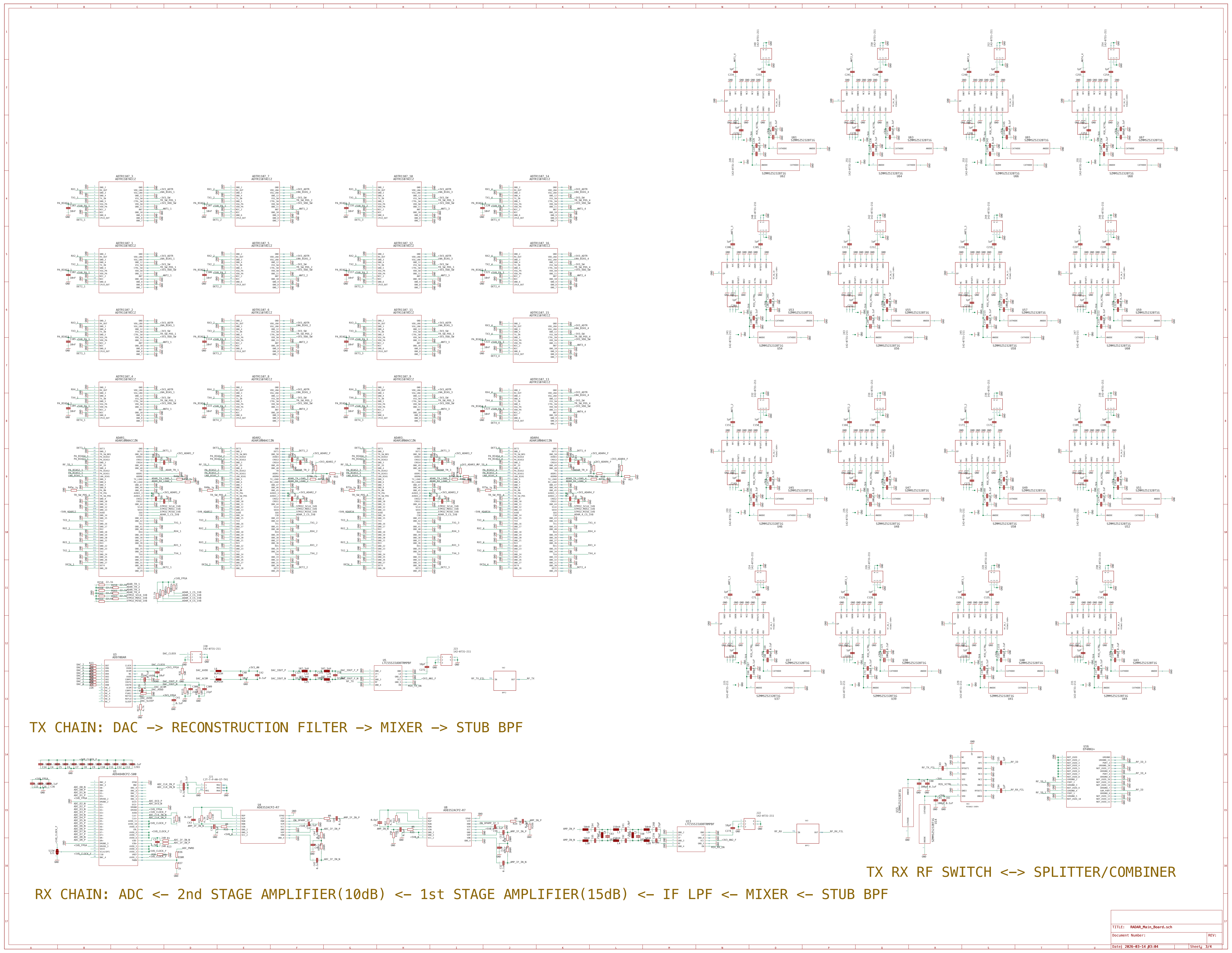
ADTR1107_3 ADTR1107ACCZ
ADTR1107_7 ADTR1107ACCZ
ADTR1107_10 ADTR1107ACCZ
ADTR1107_14 ADTR1107ACCZ
ADTR1107_1 ADTR1107ACCZ
ADTR1107_5 ADTR1107ACCZ
ADTR1107_12 ADTR1107ACCZ
ADTR1107_16 ADTR1107ACCZ
ADTR1107_2 ADTR1107ACCZ
ADTR1107_6 ADTR1107ACCZ
ADTR1107_11 ADTR1107ACCZ
ADTR1107_15 ADTR1107ACCZ
ADTR1107_4 ADTR1107ACCZ
ADTR1107_8 ADTR1107ACCZ
ADTR1107_9 ADTR1107ACCZ
ADTR1107_13 ADTR1107ACCZ
ADAR1 ADAR1000ACCZN
ADAR2 ADAR1000ACCZN
ADAR3 ADAR1000ACCZN
ADAR4 ADAR1000ACCZN
U3 AD9708AR
J20 142-0731-211
U5 LTC5552IUDBTRMPBF
J23 142-0731-211
BPF3
RF_TX
U1 AD9484BCPZ-500
J21 CJT-T-P-HH-ST-TH1
U4 AD8352ACPZ-R7
U8 AD8352ACPZ-R7
U13 LTC5552IUDBTRMPBF
J22 142-0731-211
BPF2
RF_RX
RF_RX_FIL
U61 SZMMSZ5232BT1G
U62 SZMMSZ5232BT1G
U63 SZMMSZ5232BT1G
U64 SZMMSZ5232BT1G
U65 SZMMSZ5232BT1G
U66 SZMMSZ5232BT1G
U67 SZMMSZ5232BT1G
U68 SZMMSZ5232BT1G
U53 SZMMSZ5232BT1G
U54 SZMMSZ5232BT1G
U55 SZMMSZ5232BT1G
U56 SZMMSZ5232BT1G
U57 SZMMSZ5232BT1G
U58 SZMMSZ5232BT1G
U59 SZMMSZ5232BT1G
U60 SZMMSZ5232BT1G
U45 SZMMSZ5232BT1G
U46 SZMMSZ5232BT1G
U47 SZMMSZ5232BT1G
U48 SZMMSZ5232BT1G
U49 SZMMSZ5232BT1G
U50 SZMMSZ5232BT1G
U51 SZMMSZ5232BT1G
U52 SZMMSZ5232BT1G
U17 SZMMSZ5232BT1G
U37 SZMMSZ5232BT1G
U38 SZMMSZ5232BT1G
U39 SZMMSZ5232BT1G
U40 SZMMSZ5232BT1G
U41 SZMMSZ5232BT1G
U43 SZMMSZ5232BT1G
U44 SZMMSZ5232BT1G
U15 SZMMSZ5232BT1G
U14 SZMMSZ5232BT1G
U16 EP4RKU+
TX CHAIN: DAC -> RECONSTRUCTION FILTER -> MIXER -> STUB BPF
RX CHAIN: ADC <- 2nd STAGE AMPLIFIER(10dB) <- 1st STAGE AMPLIFIER(15dB) <- IF LPF <- MIXER <- STUB BPF
TX RX RF SWITCH <-> SPLITTER/COMBINER
TITLE: RADAR_Main_Board.sch
Document Number:
REV:
Date: 2026-03-14 03:04
Sheet: 3/4

AERIS-10 — MAIN_BOARD — Schematic sheet 4/4

Drawing ID: ELEC-SCH-MB-S4 Revision: A Date: 2026-10-09 Units: mm Scale: 1:1 at the stated page size
Source: 4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.sch (EAGLE 7.4.0 XML; file revision: none recorded in the source) Status: SOURCE-DERIVED
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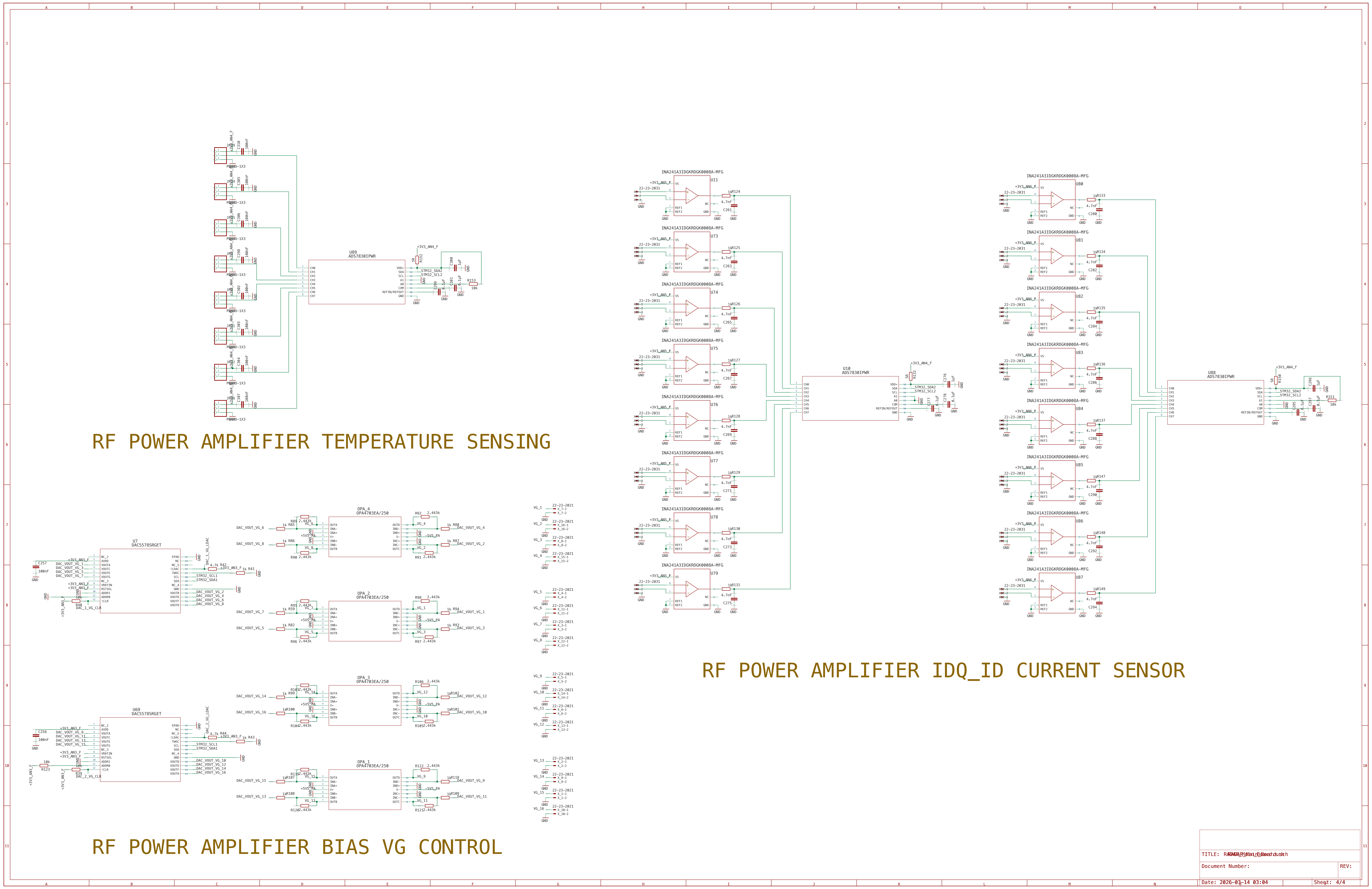
RF POWER AMPLIFIER TEMPERATURE SENSING
JP19
JP14
JP15
JP9
JP8
JP11
JP12
JP16
PGNBD-1X3
+3V3_AN4_F
C310
C305
C306
C298
C302
C303
C304
C307
100nF
GND
U89
ADS7830IPWR
CH0
CH1
CH2
CH3
CH4
CH5
CH6
CH7
VDD+
SDA
SCL
A1
A0
COM
REFIN/REFOUT
STM32_SDA2
STM32_SCL2
5R
R152
C300
1µF
C299
0.1µF
C301
R153
10k
RF POWER AMPLIFIER IDQ_ID CURRENT SENSOR
INA241A3IDGKRDGK0008A-MFG
U11
U73
U74
U75
U76
U77
U78
U79
U80
U81
U82
U83
U84
U85
U86
U87
+3V3_AN5_F
+3V3_AN6_F
22-23-2031
VS
REF1
REF2
NC
1k
R124
R125
R126
R127
R128
R129
R130
R131
R133
R134
R135
R136
R137
R147
R148
R149
4.7nF
C261
C263
C265
C267
C269
C271
C273
C275
C280
C282
C284
C286
C288
C290
C292
C294
U10
ADS7830IPWR
R132
C276
C277
C278
U88
ADS7830IPWR
R150
R151
C295
C296
C297
RF POWER AMPLIFIER BIAS VG CONTROL
U7
DAC5578SRGET
U69
DAC5578SRGET
NC_2
AVDD
VOUTA
VOUTC
VOUTE
VOUTG
NC_3
VREFIN
RSTSEL
ADDR1
ADDR0
!CLR
EPAD
NC_5
!LDAC
TWOC
NC_4
VOUTB
VOUTD
VOUTF
VOUTH
+3V3_AN3_F
C257
C256
R40
R39
R123
DAC_1_VG_CLR
DAC_2_VG_CLR
DAC_1_VG_LDAC
DAC_2_VG_LDAC
4.7k R42
4.7k R44
1k R41
1k R43
STM32_SCL1
STM32_SDA1
DAC_VOUT_VG_1 … DAC_VOUT_VG_16
OPA_1
OPA_2
OPA_3
OPA_4
OPA4703EA/250
OUTA
INA-
INA+
V+
INB+
INB-
OUTB
OUTD
IND-
IND+
V-
INC+
INC-
OUTC
+5V5_PA
-5V5_PA
2.443k
R85
R86
R87
R88
R89
R90
R91
R92
R59
R82
R93
R94
R95
R96
R97
R98
R99
R100
R101
R102
R103
R104
R105
R106
R107
R108
R109
R118
R119
R120
R121
R122
VG_1 … VG_16
22-23-2021
X_7-1
X_7-2
X_16-1
X_16-2
X_8-1
X_8-2
X_15-1
X_15-2
X_4-1
X_4-2
X_11-1
X_11-2
X_3-1
X_3-2
X_12-1
X_12-2
X_5-1
X_5-2
X_14-1
X_14-2
X_6-1
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X_13-1
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X_10-1
X_10-2
TITLE: RADAR_Main_Board.sch
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Date: 2026-03-14 03:04
Sheet: 4/4